

Seafloor spreading

Behind the volcanic arc, there is an active spreading ridge where the Philippine and Mariana plates are moving away from each other. Here, intense volcanic activity creates new oceanic crust as magma rises from the underlying mantle.

Mariana Microplate

Active arc

Most volcanoes of the active volcanic arc are seamounts, but some have risen enough to form islands.

Volcanic activity

As the Pacific Plate sinks deep into the mantle, it releases water that triggers melting in the overlying rocks.

Forearc

The region between the volcanic arc and trench is called the forearc. Here, the forces of subduction are evident in the numerous faults that slice up the forearc rock sequences.

Pacific Plate

Faults in the Pacific Plate enable it to bend and plunge almost vertically beneath the Philippine Plate.

The **pressure** at the bottom of the **Challenger Deep** is **equal** to that underneath **220 Empire State Buildings** stacked on top of each other

**△ ARC ISLAND**

The Mariana island arc parallels the curve of the Mariana Trench. Volcanic island arcs like this are created by the eruption of magma along a line roughly equidistant from the trench, which over time builds up the seafloor to create volcanic structures that rise above the waves. The island of Guam was formed by the union of two volcanoes.

Guyot

A guyot is a seamount that was once above sea level, where wave erosion planed off its top, creating a flat surface. As the seabed subsided, the guyot sank beneath the waves.

Mariana Trench

The sinking Pacific Plate pulls the overlying seafloor down with it, creating a deep trench, or groove, at the bottom of the ocean.

Pacific Plate movement

Both the Pacific and Philippine plates are moving in a northwesterly direction relative to the underlying asthenosphere, but the Pacific Plate is moving faster.

**COLORFUL FISH**

Small, brightly colored fish such as this groppo are common at depths of 1,150–1,640 ft (350–500 m) around the seamounts closest to the trench.